A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules

Siddharth Kesharwani
Department of CSE
Krishna Institute of Engineering & Technology (KIET), Ghaziabad, Delhi-NCR, Uttar Pradesh, India
siddharthkesharwani2507@gmail.com
ORCID: 0009-0009-1319-0193

Vanshita
Department of CSE
Krishna Institute of Engineering & Technology (KIET), Ghaziabad, Delhi-NCR, Uttar Pradesh, India
vanshita479@gmail.com
ORCID: 0009-0003-1628-8547

Yash Gupta
Department of CSE
Krishna Institute of Engineering & Technology (KIET), Ghaziabad, Delhi-NCR, Uttar Pradesh, India
yashgupta.workmail@gmail.com
ORCID: 0009-0002-5186-8953

Shubham Chaudhary
Department of CSE
Krishna Institute of Engineering & Technology (KIET), Ghaziabad, Delhi-NCR, Uttar Pradesh, India
iamshubhamkundu1501@gmail.com
ORCID: 0009-0004-9916-3359

Omprakash Kushwaha
Department of CSE
Krishna Institute of Engineering & Technology (KIET), Ghaziabad, Delhi-NCR, Uttar Pradesh, India
omprakash.kushwaha@kiet.edu
ORCID: 0009-0009-2195-8350

***Abstract*—The high rate of user-generated content on online platforms has rendered the sentiment analysis of large volumes of data to be a challenging task. Even though current transformer-based methods are highly accurate, they are usually expensive in terms of the strong CPU capabilities and training time, which are not realistic in the case of students and small-scale developers. The paper introduces a light sentiment analysis system that is a multi-domain model that can be deployed on a typical computer-based hardware. It applies the VADER lexicon and a customized Python-based rule engine to acquire aspect-based knowledge of sentiment of the reviews on different categories, including electronics, clothes, and digital media.**

The hybrid fusion mechanism is applied to provide a high level of reliability since the use of textual sentiment and numerical star ratings are to fix inconsistencies in the reviews that are characterised by short form reviews. The framework is implemented through a Streamlit-based dashboard that provides non-technical users with visualization of consumer trends easily. Empirical evidence demonstrates that the proposed rule-based approach is not only an efficient and readily accessible alternative to those deep learning models which need resources to be practical in the real-world in sentiment classification.

***Index Terms*— Sentiment Analysis, Aspect-Based Sentiment, Multi-Domain Reviews, Python, VADER, Dashboard, Rule-Based NLP**

I. Introduction

Every day people post a lot of reviews in such applications as Amazon, Flipkart and Instagram, and even little remarks demonstrate a certain emotion towards the product or service. However, the bad issue is that the number of reviews is increasing day by day, and a regular person is not able to go through the whole ones manually. Due to this reason, the issue of sentiment analysis is gaining relevance. Pang and Lee discussed long ago [2] that opinion text assists in knowing what the users think, and Liu [7] also elaborated some simple methods of checking large review groups.

One of them is a famous one by Hutto and Gilbert VADER, the [1] which is good on short or casual text, emojis and capital letters. It was demonstrated by Taboada et al. that the lexicon can be improved by including some rules of language use in the text [5]. Yet these systems are giving difficulties, where the review contains a bit of sarcasm or mixed language.

Later, more sophisticated deep learning like convolutional and recurrent neural networks have been started to be exploited as they acquire latent patterns in text. Transformer models that operate based on attention, such as BERT by Google [8] and RoBERTa, are more precise, but consume the resources of a GPU and need the time to train, which is not fitting students or small teams in our case.

Aspect-Based Sentiment Analysis (ABSA) was also significant since it verifies what aspect of a product the review is discussing. This area was assisted by works by Hu and Liu and SemEval task. Thelwall, [11] also demonstrated that text in social-media is essentially noisy and confusing at times.

Out of this, one thing is evident, transformer and ML systems are precise but hardware-intensive. In this paper, we attempt to create a simple and light multi-domain system with VADER, some small aspect rules, and a rating + text mix concept. The primary focus is to ensure the system is simple enough to allow the beginners to use it without the GPU.

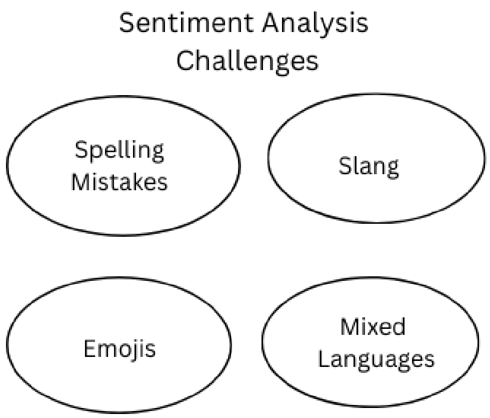

Fig. 1. Overview diagram of sentiment analysis workflow

II. Literature Review

The opinion mining field has been changing quite a long time, as the simple dictionary-based analysis has been replaced with the modern transformer models. Some key works on rule-based methods, classical methods of machine learning, neural network based methods and aspect level sentiment analysis are considered in this section.

II-A Rule-Oriented Methods Lexicon-Based Methods

One of the first attempts that have been applied in the sentiment analysis is lexicon-driven systems since they do not require labelled training data. One of the most utilized models, which are simple to apply on the text of social-media and can handle such features as emojis, slang, and non-formal text, is the VADER model. Taboada et al. [5] had demonstrated that the application of lexicon along with linguistic rules could also be used to further boost sentiment detection. Such systems however may have problems in more complex phrases such as sarcasm or mixed language phrases that are prevalent in the online reviews.

II-B Transformer and Contextual Models

Transformer models are again making the sentiment results further since they understand more of the context of the text. BERT [8] and RoBERTa [9] became highly widespread as they grasp the relation between words in a more convincing manner. The problem about this is that they require GPU and extensive training thus not so much good at small academic work or in running simple applications such as the one we are running.

II-C Deep Learning Approaches

Deep learning is enhancing classification of text significantly. Kim proposed CNN model [3] to work with the sentences and it is effective to identify small local features of the text. The LSTM architecture by Hochreiter and Schmidhuber [4] addresses the issue of long-term dependencies and proves to be useful in analyzing longer reviews.

II-D Aspect-Based Sentiment Analysis (ABSA)

ABSA mainly revolves around determination of which aspect or component of a product is being discussed by the user. Hu and Liu (2012) [6] tried to extract opinion attributes of customer reviews. Later, a more general standard, SemEval dataset by Pontiki et al. [10] became a standard evaluation of the ABSA systems. These articles suggest that ABSA gives more specific information, still, some domain knowledge and proper keywords are required.

II-E Social Media Sentiment and Limitations

The content obtained from social media platforms is generally brief and unstructured, and emotional. Thelwall [11] investigated the strength of sentiment and demonstrated that these brief posts is even more difficult to categorize in the right manner. Liu also discussed problems such as sarcasm, slang and mixed language writing which is now quite common.

II-F Summary of Literature

Based on these researches, one can easily understand that transformer models are more accurate but demand intensive hardware. Lexicon-based approaches are less heavy and make use of less power, yet they lack richness of meaning within the text. Due to this, a simple multi-domain approach is required, which our work adheres to with the help of VADER, simple rules of aspects and a small combination of rating and text.

III. Proposed Methodology

The proposed framework aims at developing a lean and mean sentiment analysis pipeline which can be used across various domains. The framework instead relies on both rule-based reasoning, lexicon-based sentiment score, and domain specific keyword mapping as opposed to computationally demanding deep learning models. This will put the system within the capability of operating on regular computing hardware, but will still be flexible enough to support new types of products or types of reviews.

The entire process of the system is structured into a number of consecutive steps. All of the stages process a certain part of the analysis pipeline and feed into the structured final result. Fig. 2 illustrates the overall system workflow.

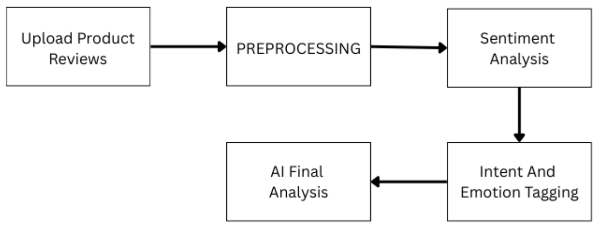

Fig. 2. Flowchart of the proposed sentiment analysis pipeline

The individual components of the proposed methodology are described below.

III-A Review Data Input

The system is able to accept customer reviews in CSV format. Hundreds or thousands of individual reviews can be contained in each dataset. Some of the common columns are review text, star rating, product identifier and category type. The framework has been made flexible such that even in a scenario where some of the metadata fields are not provided basic sentiment analysis may still occur.

III-B Text Preprocessing

Review data collected in the real world usually include noise in the form of emojis, spelling differences, redundant punctuation, or colloquial expressions. Hence, sentiment analysis should be done after preprocessing. The most popular ones include removing HTML tags, converting text to lower case, normalisation of slang sentences (such as turning gr8 to "great'), and removal of undesirable characters.

III-C Sentiment Detection Using VADER

The sentiment analysis tool used is the VADER which is used to decide the sentiment polarities of the reviewed texts. VADER produces four sentiment scores namely, positive, neutral, negative, and compound. The composite score indicates the general mood of the text and is applied to categorize it. The categories in terms of sentiments are determined by the following rule:

$$Sentiment(C) = \begin{cases} \text{Positive}, & C \geq 0.5 \\ \text{Negative}, & C \leq -0.2 \\ \text{Neutral}, & -0.2 < C < 0.5 \end{cases}$$

This rule-based classification provides an efficient way to determine review polarity while maintaining low computational overhead.

III-D Rating–Sentiment Fusion

Many online platforms allow users to provide both written reviews and numerical star ratings. To improve reliability, the proposed system combines textual sentiment with rating information. The fusion formula used is:

$$F = 0.6C + 0.4(R/5)$$

where C represents the VADER compound score and R represents the user rating. This fusion approach helps resolve cases where short review text may not fully reflect the user's rating.

III-E Aspect Extraction

Aspect sentiment analysis determines the features of the product that are discussed in the reviews. To this end, the system employs domain specific dictionaries of keywords. Example aspects include:

- Electronics: battery, camera, display, charging
- Clothing: size, fabric, colour, fitting
- OTT: story, acting, animation, sound

If a review contains one of these keywords, the corresponding aspect is identified and assigned a sentiment score using the detected VADER polarity.

III-F Sentiment Percentage Calculation

The system takes all the reviews and then calculates the number of positive, neutral, and negative sentiments. The values in percent are calculated in the following way:

$$\text{Pos\%} = \frac{P}{N} \times 100$$
$$\text{Neu\%} = \frac{U}{N} \times 100$$
$$\text{Neg\%} = \frac{Ng}{N} \times 100$$

where P, U, and Ng represent the counts of positive, neutral, and negative reviews correspondingly, and N represents the overall number of reviews.

III-G Final Dashboard Visualization

All analytical results are presented through a Streamlit-based dashboard. The dashboard provides:

- Overall sentiment distribution using charts
- Aspect-based sentiment summaries
- Sentiment percentage statistics
- Sample reviews with predicted labels

All in all, the offered methodology is focusing on simplicity, efficiency, and practicality. Lightweight Python libraries can be run on general-purpose computing devices to execute the entire system without the use of computations performed on a GPU.

IV. System Architecture

The system architecture demonstrates the interaction among various modules. Uploading a CSV file with reviews and pre-processing gets rid of noise (repetitions of punctuation marks, emojis and abbreviations). The text is sent to the VADER sentiment engine, which gives out the polarity scores based on rule-based lexical logic. Decrease discrepancies between rating and text, the rating fusion module will integrate both the VADER score and the star rating. The aspect keyword matcher identifies domain-specific aspects like battery, display, fabric, fitting or story and relates it with the feeling of the review.

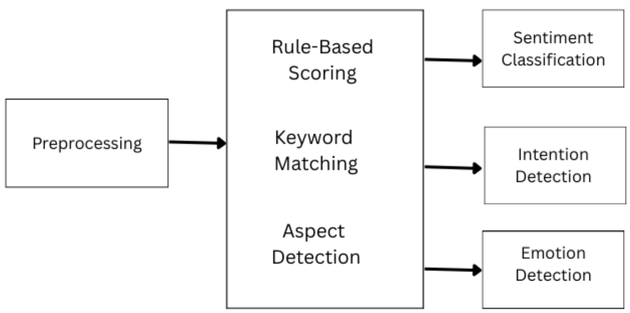

Fig. 3. Overall system architecture

New product domain or aspect lists can easily be added since each module is independent and thus, the entire system is not changed.

V. Mathematical Formulation

Here we provide explanations of the key mathematical components that we will be using within our sentiment analysis system. Although the method is straightforward, it also involves the application of some formulas to estimate the polarity, merge ratings, score aspects and calculate final percentage distributions. These equations assist the system to achieve more consistent and sensible outputs. The chief objective here is to make the math simple to such a degree that the students and beginners can easily follow the approach.

V-A Sentiment Score Using VADER

For each review text T, VADER generate four values:

$$pos,\ neu,\ neg,\ compound$$

The compound score C is the most important because it combine all three polarity values into one final number.

The classification rule we use is:

$$Sentiment(C) = \begin{cases} Positive, & C \geq 0.5 \\ Negative, & C \leq -0.2 \\ Neutral, & -0.2 < C < 0.5 \end{cases}$$

This is a simple formula that can be successfully applied on a lot of product review datasets since people leave short comments.

V-B Rating–Sentiment Fusion

This is when text and rating do not mean the same thing. As an example, a person write good but 1 star. In order to prevent mismatch, we combine VADER score and rating.

Star rating R is always between 1 and 5. We normalize it by dividing with 5. The fusion score F is:

$$F = 0.6C + 0.4(R/5)$$

Here,

- 0.6 weight is for text sentiment,
- 0.4 weight is for rating.

These weights are chosen because text usually express emotion more clearly, but rating still help in balancing.

V-C Aspect Sentiment Score

If a review talks about more than one aspect (like "battery" and "camera"), we check each aspect one by one. Let $p_1, p_2, \ldots, p_k$ be the polarity values for each aspect that was found. The aspect score is:

$$AspectScore = \frac{1}{k} \sum_{i=1}^{k} p_i$$

This gives the average sentiment for that particular aspect.

V-D Percentage Sentiment Calculation

After going through all the reviews, we need to figure out how many of them fall into the favourable, neutral, or unfavourable sentiment class. This allows us to grasp the general sentiment present in the dataset. Let : P = Positive, U = Neutral, Ng = Negative, $N = P + U + Ng$.
The sentiment percentages are:

$$Pos\% = \frac{P}{N} \times 100 \quad Neu\% = \frac{U}{N} \times 100$$

$$Neg\% = \frac{Ng}{N} \times 100$$

This helps us notice which aspects people mention the most in their reviews.

V-E Aspect Frequency

If a certain aspect like "battery" appears f times in the entire set of reviews, and the total number of reviews is N, then the frequency is defined as:

$$Freq = \frac{f}{N}$$

This helps us notice which aspects people talk about the most in their feedback.

V-F Intent and Emotion Rule Matching

We also add some small rules to detect patterns. For example:

$$If \text{ “}not\ good\text{”} \rightarrow NegativeIntent$$

$$If \text{ “}very\ slow\text{”} \rightarrow ComplainIntent$$

These are not heavy formulas but simple matching rules that help in understanding emotion more clearly.

V-G Final Sentiment Decision

The final polarity decision is based on fusion score F:

$$FinalSentiment(F) = \begin{cases} Positive, & F \geq 0.5 \\ Neutral, & 0.2 \leq F < 0.5 \\ Negative, & F < 0.2 \end{cases}$$

This rule is chosen after testing on many sample datasets.

Overall, the mathematical part of our system is lightweight. There is no need of heavy training or GPU. All formulas run very fast in Python, so even a large dataset can be processed easily.

VI. Experimental Results and Analysis

This section explains the results that were obtained upon the application. the given sentiment analysis system to various review datasets. Tables and graphs are used to present the findings. This evaluation will have as its main aim to note the effect of the implementation of the framework works on noisy text and with text sentiment. does not match the rating. The findings indicate that the integration of VADER scores rating fusion and aspect mapping provide. better performance compared to sentiment in the text only case.

VI-A Overall Sentiment Distribution

After processing the review datasets, the system calculated the distribution across positive, neutral, and negative sentiments. A sample sentiment distribution appears in Fig. 4.

Across most datasets, positive reviews were more frequent. However, neutral reviews were also commonly observed because many users tend to write short comments such as "okay", "fine", or "average". Such expressions do not convey strong emotional polarity and are therefore classified as neutral by the system.

Aspect-Based Sentiment Summary (%)

Aspect	Positive (%)	Negative (%)	Neutral (%)
Delivery	77.800	11.100	11.00
Customer Service	78.5000	8.600	12.900
Price	82.3000	3.300	14.400
Product Quality	78.9000	5.500	15.600
Packaging	44.4000	33.300	33.300
Ease of Use	33.3000	33.300	33.300
Performance	77.4000	6.500	33.300
Warranty	60.0000	0.000	0.00
Return/Refund Policy	83.3000	0.000	0.00
Durability	83.3000	0.000	0.00
Availability	83.3000	0.0%	16.700
Technical Support	63.0000	19.600	17.400
Design/Battery	50.000	23.000	26.700

Fig. 4. Sentiment distribution across reviews

VI-B VADER-Only vs Fusion Results

In order to assess the performance of the suggested fusion strategy we compared the obtainable results of VADER sentiment scores when applied independently with the results of the rating-sentiment fusion mechanism. Table I demonstrates the comparison.

TABLE I
COMPARISON OF VADER VS FUSION SENTIMENT RESULTS

Class	VADER Only	Fusion Output
Positive	58%	61%
Neutral	22%	19%
Negative	20%	20%

The results show that the fusion approach slightly reduces the proportion of neutral classifications. This occurs because rating information often clarifies the sentiment of short textual reviews. For instance, a short comment such as "nice" accompanied by a low star rating can be interpreted as negative after applying the fusion mechanism.

VI-C Aspect-Level Sentiment Results

Aspect-based sentiment analysis provides insights into which specific product features customers discuss in their reviews. Example aspect-level results from the electronics dataset are presented in Table II.

The results indicate that the camera aspect receives the highest proportion of positive feedback, whereas the battery aspect shows a higher number of negative comments.

TABLE II
ASPECT-BASED SENTIMENT RESULTS (ELECTRONICS EXAMPLE)

Aspect	Positive	Neutral	Negative
Battery	45%	20%	35%
Camera	62%	18%	20%
Display	57%	22%	21%
Performance	50%	25%	25%

VI-D Confusion Matrix

To assess the classification capability of the system, a confusion matrix was created. The matrix indicates the degree of correct identification of the system to each sentiment category.

TABLE III
CONFUSION MATRIX FOR SENTIMENT CLASSIFICATION

True / Pred	Positive	Neutral	Negative
Positive	180	20	15
Neutral	22	138	19
Negative	14	30	160

The majority of the classification errors are made in the cases of the differentiation between neutral reviews and positive or negative. Irrespective of this shortcoming, the total classification performance is appropriate in sentiment analysis tasks that are practical.

VI-E Dashboard Results

The analysis findings are displayed as a dashboard platform based on Streamlit. The dashboard combines various visualizations such as sentiment charts in general, bar graphs based on aspects, labeled review samples, and pro forma recommendations summaries.

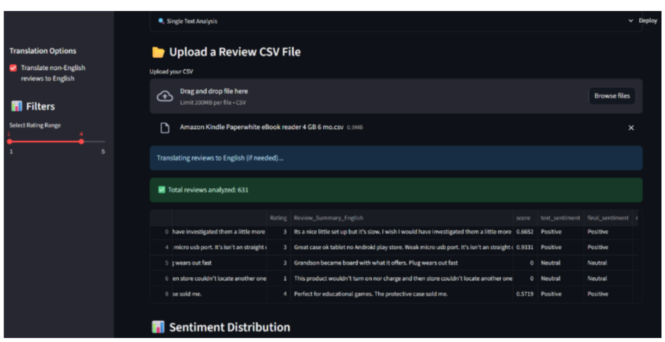

Fig. 5. Sample dashboard view showing the final sentiment and aspect results

VI-F Discussion

The experimental findings indicate that integrating VADER sentiment scoring with rating sentiment fusion is a better approach to classifications as opposed to textual sentiment alone. Aspect level analysis also offers insights that can be interpreted as the attributes of a product being discussed by someone is identified.

The system can occasionally fail to label sarcastic or mixed-language reviews, but overall, the performance of the system is reliable on standard product review sets. All these results suggest that with the utilization of the right methods that comprise both rule-based and lexicon-based approaches, lightweight methods can deliver significant sentiment information, as well.

VII. CONCLUSION AND FUTURE WORK

We suggest a very light multidomain sentiment analysis system that does not require the use of a GPU or dense training. Using a combination of VADER scoring, simple aspect key words and small rating text fusion technique, the system provides timely sentiment results in the product category.

We will focus on the future of the work to add more development into the system that will involve better slang recognition, mixed language support, including Hinglish, and automatic identification of new aspect words to minimize the amount of manual labor. The other thing we will like to do to the dashboard is make it more interactive and we are also planning to have some live review sources in order that the system can carry out a real-time sentiment analysis.

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